

SQL for Data Science

1. What is SQL and Why Data Scientists Use It

SQL (Structured Query Language) is used to retrieve, filter, aggregate, and combine data stored in relational databases.

In data science, SQL is mainly used to:

- Extract data from databases
- Perform filtering and aggregation
- Join multiple tables
- Prepare clean datasets for analysis in Python, Power BI, or Tableau

SQL is usually the first step before Python or visualization tools.

2. Basic Structure of an SQL Query

```
SELECT column_name  
FROM table_name  
WHERE condition;
```

Execution order (important for understanding queries):

1. FROM
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT
6. ORDER BY

3. SELECT Statement

Used to choose columns from a table.

```
SELECT name, age, salary  
FROM employees;
```

Select all columns:

```
SELECT *  
FROM employees;
```

In data science, avoid `SELECT *` for large tables.

4. WHERE Clause (Filtering Rows)

Used to filter records based on conditions.

```
SELECT *  
FROM employees  
WHERE age > 30;
```

Common operators:

- `=, !=`
- `>, <, >=, <=`
- `AND, OR, NOT`

Example:

```
SELECT *  
FROM employees
```

```
WHERE department = 'IT' AND salary > 50000;
```

5. ORDER BY (Sorting Data)

Used to sort results.

```
SELECT name, salary  
FROM employees  
ORDER BY salary DESC;
```

Default is ascending order.

6. Aggregate Functions (Very Important)

Used to summarize data.

Function	Purpose
COUNT	Number of rows
SUM	Total
AVG	Average
MIN	Minimum
MAX	Maximum

Example:

```
SELECT AVG(salary)  
FROM employees;
```

7. GROUP BY (Core for Analytics)

Used with aggregate functions to group data.

```
SELECT department, AVG(salary)
FROM employees
GROUP BY department;
```

Rules:

- Every column in SELECT must be either:
 - Aggregated
 - Present in GROUP BY

8. HAVING Clause

Filters grouped data (used after GROUP BY).

```
SELECT department, AVG(salary)
FROM employees
GROUP BY department
HAVING AVG(salary) > 60000;
```

Difference:

- WHERE → filters rows
- HAVING → filters groups

9. JOINS (Extremely Important for Data Science)

Used to combine data from multiple tables.

9.1 INNER JOIN

Returns matching records from both tables.

```
SELECT e.name, d.department_name
FROM employees e
INNER JOIN departments d
ON e.department_id = d.department_id;
```

9.2 LEFT JOIN

Returns all rows from left table and matching rows from right table.

```
SELECT e.name, d.department_name
FROM employees e
LEFT JOIN departments d
ON e.department_id = d.department_id;
```

Most commonly used join in analytics.

10. Handling Missing Values (NULL)

Check for missing values:

```
SELECT *
FROM employees
WHERE salary IS NULL;
```

Replace NULL values:

```
SELECT name, COALESCE(salary, 0)
FROM employees;
```

11. CASE Statement (Conditional Logic)

Used to create categories.

```
SELECT name,  
       CASE  
           WHEN salary >= 80000 THEN 'High'  
           WHEN salary >= 50000 THEN 'Medium'  
           ELSE 'Low'  
       END AS salary_category  
FROM employees;
```

Very useful for feature engineering.

12. Subqueries

A query inside another query.

```
SELECT *  
FROM employees  
WHERE salary > (  
    SELECT AVG(salary)  
    FROM employees  
);
```

Used for:

- Filtering based on aggregated values
- Advanced analysis logic

13. Window Functions (Intermediate but Valuable)

Used for calculations without collapsing rows.

Example: Ranking salaries.

```
SELECT name, salary,  
       RANK() OVER (ORDER BY salary DESC) AS salary_rank
```

FROM employees;

Common window functions:

- **ROW_NUMBER()**
- **RANK()**
- **DENSE_RANK()**

14. SQL for Data Science Workflow

Typical flow:

- 1. Connect to database**
- 2. Use SQL to:**
 - **Filter rows**
 - **Join tables**
 - **Aggregate data**
- 3. Export result to:**
 - **Python**
 - **Power BI**
 - **Tableau**

SQL reduces data size before analysis.

15. SQL vs Python in Data Science

Task	SQL	Python
Data extraction	Yes	Limited
Filtering	Yes	Yes
Aggregation	Yes	Yes
Visualization	No	Yes
Machine learning	No	Yes

SQL prepares the data.

Python analyzes the data.